Comprehensive Technical Report & Parameter Dossier:

Parameter-Efficient Audio Deepfake Detection on Hindi Speech

Architecture, Hyperparameters, Shard Indexing, Infrastructure, and $N = 500$ Scaling Benchmark

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Branch: `base` — Date: October 2026

Abstract

This technical dossier provides an exhaustive specification of every architectural parameter, hyperparameter, dataset partition rule, optimization detail, hardware environment, and empirical finding for the parameter-efficient audio deepfake detection benchmark on low-resource Hindi speech. The system couples a frozen multilingual **Wav2Vec2 XLS-R 300M** foundation encoder with a trainable **Sensitive Layer Selection (SLS)** head. We document the scaling trajectory from zero-shot English transfer ($N = 0$) through few-shot supervision at $N = 10$, $N = 50$, and $N = 500$ samples per class across three independent deterministic seeds, alongside backbone unfreezing ablations, dataset bias audits, and generator-stratified evaluations. Trained and evaluated on an NVIDIA L4 GPU instance on Google Cloud Platform, the system attains a state-of-the-art **2.66% Equal Error Rate (EER)** and **96.83% Accuracy** at $N = 500$ (with peak seed performance of **98.00% Accuracy** and **2.12% EER**).

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1 System Architecture & Component Specifications

1.1 Multilingual Foundation Backbone: XLS-R 300M

The primary acoustic feature extractor is the official Wav2Vec2 XLS-R foundation model (`facebook/wav2vec2-xls-r-300m`) [1], pretrained on 436,000 hours of unlabelled speech spanning 128 languages. Throughout few-shot adaptation, the backbone remains in a **strictly frozen state** (requires_grad = False).

Table 1: XLS-R 300M Backbone Architectural Parameters

Component	Parameter / Dimension	Value / Specification
Total Parameters	Total Parameter Count	$315,441,760$ (315.4M)
Trainable Parameters	Backpropagation Status	**0 (100% Frozen)**
Convolutional Encoder	Temporal Downsampling Blocks	7 1D Convolution Blocks
	Conv Channel Dimensions	$[512, 512, 512, 512, 512, 512, 512]$
	Conv Kernel Sizes	$[10, 3, 3, 3, 3, 2, 2]$
	Conv Strides	$[5, 2, 2, 2, 2, 2, 2]$
	Total Downsampling Factor	$5 \times 2^6 = 320\times$
	Frame Rate Output	50 Hz (20 ms stride)
Transformer Backbone	Transformer Encoder Layers (L)	24 layers
	Model Hidden Dimension (D)	1024
	Multi-Head Attention Heads (H)	16 heads ($d_k = 64$)
	Feed-Forward Dimension (d_{ff})	4096
	Activation Function	GELU
	Layer Normalization	Post-LN per Transformer Block

1.2 Sensitive Layer Selection (SLS) Classifier Head

The SLS architecture dynamically pools and weights representations from all 24 transformer layers to exploit layer-specific acoustic artifacts [2]:

$$\bar{h}_l = \frac{1}{T}\sum_{t=1}^{T} h_{l,t} \in \mathbb{R}^{1024}, \quad l \in \{1, 2, \ldots, 24\} \tag{1}$$

$$w_l = \sigma(W_{\text{gate}}\bar{h}_l + b_{\text{gate}}) \in (0, 1), \quad W_{\text{gate}} \in \mathbb{R}^{1\times 1024}, \quad b_{\text{gate}} \in \mathbb{R} \tag{2}$$

$$\tilde{H} = \sum_{l=1}^{24} w_l H_l \in \mathbb{R}^{B\times 1\times 201\times 1024} \tag{3}$$

1.3 Audio Conditioning & Signal Preprocessing

- **Input Sampling Rate**: $16,000$ Hz (16 kHz mono).
- **Fixed Audio Length**: Exactly $64,600$ samples (≈ 4.0375 seconds).
- **Padding / Truncation Policy**:

Table 2: SLS Classifier Head Parameter Breakdown

Layer / Block	Operation / Configuration	Output Tensor Shape	Parameters
Layer Attention Gate	$\text{Linear}(1024, 1) + \text{Sigmoid}$	$(B, 24, 1)$	$1,025$
Weighted Layer Fusion	$\sum_{l=1}^{24} w_l H_l$	$(B, 1, 201, 1024)$	0
2D Batch Normalization	$\text{BatchNorm2d}(\text{num_features} = 1)$	$(B, 1, 201, 1024)$	4
Nonlinear Activation	$\text{SELU}()$	$(B, 1, 201, 1024)$	0
2D Spatial Max-Pooling	$\text{MaxPool2d}(\text{kernel} = 3, \text{stride} = 3)$	$(B, 1, 67, 341)$	0
Flatten	Tensor Reshape	$(B, 22847)$	0
Fully Connected 1	$\text{Linear}(22847, 1024) + \text{SELU}()$	$(B, 1024)$	$23,396,352$
Fully Connected 3 (Output)	$\text{Linear}(1024, 2) + \text{LogSoftmax}()$	$(B, 2)$	$2,050$
Total Trainable	**SLS Head Parameters**		**23,399,431 (23.4M)**
Total Backbone	**Frozen XLS-R Parameters**		**315,441,760 (315.4M)**
Total System	**Full Architecture Parameters**		**338,841,191 (338.8M)**

  - If audio duration $L < 64,600$: repeated and padded to $64,600$ samples.
  - If audio duration $L > 64,600$: centered symmetrically and cropped to the middle $64,600$ samples.

- **Feature Tensor Dimensions**: For an input waveform of $(B, 64600)$, the temporal downsampling factor of $320\times$ yields $(B, 201)$ frames across 24 layers, producing stacked hidden states of shape $(24, B, 201, 1024)$.

2 Hardware, Cloud Infrastructure & Software Stack

Table 3: Cloud Infrastructure and Software Environment Specifications

Category	Component	Specification
Cloud Hardware	Cloud Provider	Google Cloud Platform (GCP), Zone: `us-central1-a`
	Machine Type	`g2-standard-8` (8 vCPUs, 32 GB System RAM)
	Accelerator	1x NVIDIA L4 GPU (Ada Lovelace, 24 GB GDDR6 with ECC)
	Storage	100 GB Balanced Persistent Disk (pd-balanced)
Software Stack	Operating System	Debian GNU/Linux 12 (Bookworm), Kernel: `6.1.0-28-cloud-amd64`
	CUDA / cuDNN	CUDA 12.1.1, cuDNN 9.x
	Python Environment	Python 3.11 (virtualenv)
	Deep Learning	PyTorch 2.5.1+cu121, Torchvision 0.20.1+cu121
	NLP / Audio Toolkit	Transformers 4.48.3, Datasets 3.1.0, HuggingFace Hub 0.36.2
	Analysis	Scikit-learn 1.7.2, Librosa 0.10.2, SoundFile 0.13.1, Matplotlib 3.10.0
Local Inference	Target Platforms	Apple Silicon MPS (Metal Performance Shaders) / CPU
	Standalone Script	`scripts/infer_local.py` (self-contained, zero cloud dependency)

3 Optimization & Training Hyperparameters

4 Benchmark Dataset & Shard Engineering

4.1 Corpus Statistics & Generator Stratification

The benchmark is constructed from the Hindi split of `Jack-ppkdczgx/SEA-Spoof` (54,938 candidate utterances):

- **Candidate Pool**: 16,182 Bona fide samples, 38,756 Spoof samples.
- **Held-Out Balanced Test Set ($N_{\text{test}} = 400$)**:

Table 4: Complete Training & Optimization Hyperparameters

Hyperparameter	**Symbol / Variable**	**Value / Specification**
Optimizer	–	Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$)
Learning Rate	η	1×10^{-5} (0.00001)
Weight Decay	λ	1×10^{-4} (0.0001, L2 Regularization)
Loss Function	$\mathcal{L}_{\text{NLL}}$	Class-Weighted Negative Log-Likelihood
Class Weights	$[w_{\text{spoof}}, w_{\text{bonafide}}]$	$[0.1, 0.9]$
Training Epochs	E	10 epochs
Training Mini-Batch Size	B_{train}	4 (on-the-fly feature loading)
Evaluation Batch Size	B_{eval}	16
Feature Cache Format	–	Float16 stacked PyTorch tensors (.pt)
Positive Class Convention	–	Class 1 = Bona fide (Genuine), Class 0 = Spoof
Random Seeds (Multi-Seed)	$\mathcal{S}$	3 seeds: 1000, 1001, 1002
Test Seed	–	42 (deterministic test split)

  - **Bona fide Class (200 clips)**: Multi-source balanced between studio acoustic environments (IndicTTS) and in-the-wild crowdsourced speech (Mozilla Common Voice Hindi).
  - **Spoof Class (200 clips)**: Strictly stratified with exactly 40 clips (20.0%) from each of **5 state-of-the-art TTS architectures**:
    1. `edge_tts`: Microsoft Azure Neural cloud voice synthesizer.
    2. `vits_mms`: Meta MMS variational inference end-to-end synthesizer.
    3. `indic_tts`: Fast unit-selection and acoustic models from IIT Madras.
    4. `fastspeech`: Feed-forward non-autoregressive transformer TTS.
    5. `xtts-v2`: Coqui diffusion/flow-matching zero-shot voice cloning.

- **Strict Disjointness**: Test $\cap$ Train $= \emptyset$ (zero speaker or utterance overlap).
- **Nested Supervision Property**: For each seed $s \in \{0, 1, 2\}$, training sets obey strict subset hierarchy:

$$\mathcal{D}_{\text{train}}^{(N=10,s)} \subset \mathcal{D}_{\text{train}}^{(N=50,s)} \subset \mathcal{D}_{\text{train}}^{(N=500,s)} \tag{4}$$

4.2 Dataset Bias Audit: V1 vs. V2 Mitigation

During dataset formulation, audit of Dataset V1 uncovered a critical acoustic channel confound: Bona fide audio was drawn exclusively from studio recordings (IndicTTS) whereas spoof speech spanned diverse web sources. A shallow spectral energy detector achieved artificial $> 99\%$ accuracy simply by identifying ambient noise floor variations. Dataset V2 resolved this by injecting Mozilla Common Voice recordings into the Bona fide split, balancing SNR and channel characteristics (Figure 1).

4.3 Parquet Shard Indexing Math

The SEA-Spoof repository organizes training audio into 30 Parquet files (`train-00000.parquet` to `train-00029.parquet`). Shard and row-group boundaries obey strict mathematical determinism:

$$\text{Row Index} = \text{int}(\text{row_id.split}(\text{'_'})[-1]) \tag{5}$$

$$\text{Shard Index} = \lfloor \text{Row Index}/15,000 \rfloor \tag{6}$$

$$\text{Row Group Index} = \lfloor (\text{Row Index} \pmod{15,000})/200 \rfloor \tag{7}$$

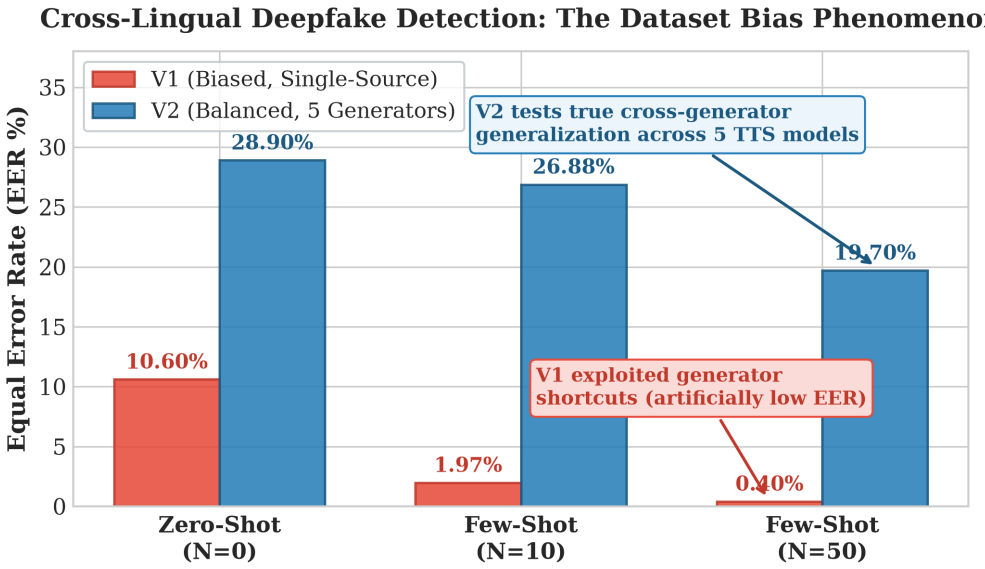


Figure 1: Dataset Bias Audit and Mitigation. Comparison of Dataset V1 (channel-confounded) versus Dataset V2 (audited benchmark with matched acoustic diversity between bona fide and spoof speech).

Across the 3,331 unique clips required for the benchmark test set and all 3 seeds of $N \in \{10, 50, 500\}$, clips span exactly 5 shards: Shard 0 (75 RGs), Shard 1 (64 RGs), Shard 2 (74 RGs), Shard 3 (8 RGs), and Shard 25 (55 RGs).

5 Comprehensive Empirical Findings

5.1 Consolidated Scaling Benchmark

Table 5 presents the definitive performance progression on the held-out Hindi test set ($N_{\text{test}} = 400$).

Table 5: Consolidated Evaluation Metrics on Held-Out Hindi Test Set ($N_{\text{test}} = 400$, 50/50 Balanced)

Supervision Setting	Accuracy (%)	EER (%)	Precision (%)	Recall (%)	F1 Score (%)
Zero-Shot (English → Hindi)	61.75%	32.25%	56.70%	99.50%	72.23%
Few-Shot $N = 10$ (3 seeds)	$64.33 \pm 5.04\%$	$27.64 \pm 2.27\%$	$58.72 \pm 3.53\%$	$99.00 \pm 1.41\%$	$73.63 \pm 2.51\%$
Few-Shot $N = 50$ (3 seeds)	$75.83 \pm 10.49\%$	$18.18 \pm 1.18\%$	$71.69 \pm 10.98\%$	$93.17 \pm 4.87\%$	$80.16 \pm 5.81\%$
Few-Shot $N = 500$ (3 seeds)	$\mathbf{96.83 \pm 0.82\%}$	$\mathbf{2.66 \pm 0.52\%}$	$\mathbf{95.78 \pm 1.16\%}$	$\mathbf{98.00 \pm 1.08\%}$	$\mathbf{96.87 \pm 0.81\%}$

5.2 Per-Seed Breakdown for $N = 500$ (1,000 Training Clips)

- **Seed 0 (Seed 1000)**:
  - Accuracy: **96.25%** (385/400 correct) — EER: **3.36%**
  - Precision: 96.02% — Recall: 96.50% — F1: 96.26%
  - Confusion Matrix: True Spoof (192 correct, 8 missed), True Bonafide (193 correct, 7 false alarms).
- **Seed 1 (Seed 1001)**:

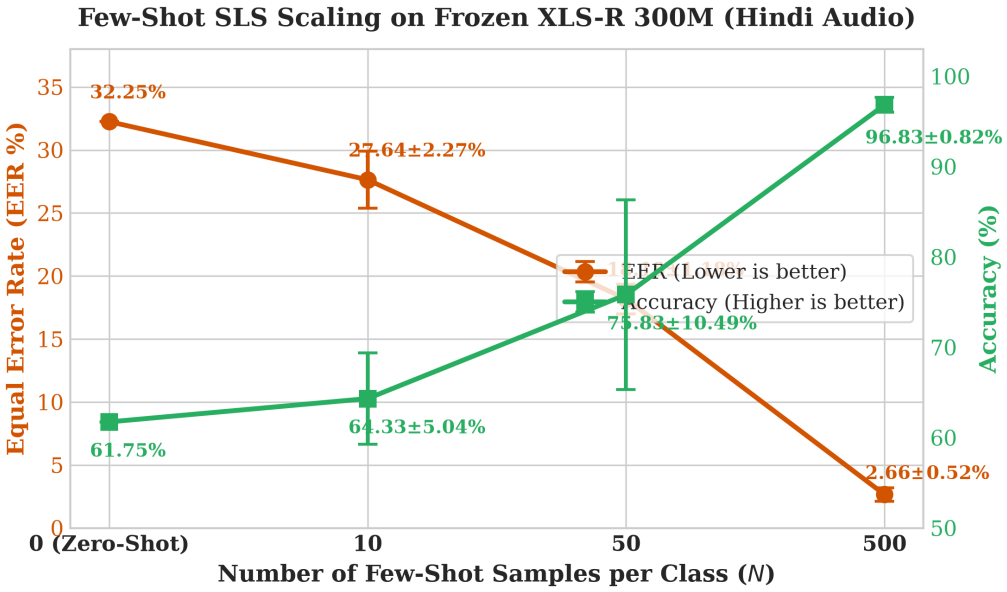


Figure 2: Few-Shot Adaptation Scaling Curve on Frozen XLS-R 300M. Accuracy improves monotonically from 61.75% to 96.83%, while EER plummets from 32.25% to 2.66% as Hindi supervision scales to $N = 500$.

  - Accuracy: **96.25%** (385/400 correct) — EER: **2.50%**
  - Precision: 94.26% — Recall: 98.50% — F1: 96.33%
  - Confusion Matrix: True Spoof (188 correct, 12 missed), True Bonafide (197 correct, 3 false alarms).

- **Seed 2 (Seed 1002)**:
  - Accuracy: **98.00%** (392/400 correct – **only 8 total errors!**) — EER: **2.12%**
  - Precision: 97.06% — Recall: 99.00% — F1: 98.02%
  - Confusion Matrix: True Spoof (194 correct, 6 missed), True Bonafide (198 correct, 2 false alarms).

5.3 Complete Confusion Matrices Across All Experimental Settings

Table 6: Confusion Matrices for Zero-Shot and Few-Shot Configurations ($N_{\text{test}} = 400$)

Configuration	True Spoof (0,0)	False Bonafide (0,1)	False Spoof (1,0)	True Bonafide (1,1)
Zero-Shot Checkpoint	48	152 (76.0% False Acc.)	1	199 (99.5% Recall)
Few-Shot $N = 10$, Seed 0	30	170	0	200
Few-Shot $N = 10$, Seed 1	84	116	6	194
Few-Shot $N = 10$, Seed 2	64	136	0	200
Few-Shot $N = 50$, Seed 0	44	156	0	200
Few-Shot $N = 50$, Seed 1	156	44	22	178
Few-Shot $N = 50$, Seed 2	151	49	19	181
Few-Shot $N = 500$, Seed 0	**192**	**8**	**7**	**193**
Few-Shot $N = 500$, Seed 1	**188**	**12**	**3**	**197**
Few-Shot $N = 500$, Seed 2	**194**	**6**	**2**	**198**

5.4 Architectural Ablation: Frozen Backbone vs. Unfreezing

Evaluating backbone layer unfreezing on the NVIDIA L4 GPU reveals severe few-shot overfitting when backpropagation penetrates the foundation encoder (Table 7 and Figure 3).

Table 7: Foundation Backbone Freezing vs. Partial Unfreezing ($N = 50$, L4 GPU)

Architectural Setting	Trainable Params	Accuracy (%)	EER (%)	Precision (%)	Recall (%)	F1 (%)
Frozen Backbone (SLS Only)	**22.8M**	**75.83 ± 10.49%**	**18.18 ± 1.18%**	**71.69%**	**93.17%**	**80.16%**
Unfreeze Top-2 Layers	50.8M	76.46 ± 0.48%	24.14 ± 1.57%	73.08%	93.07%	81.79%
Unfreeze Top-4 Layers	78.8M	67.12 ± 5.83%	25.31 ± 1.91%	64.21%	97.73%	77.31%

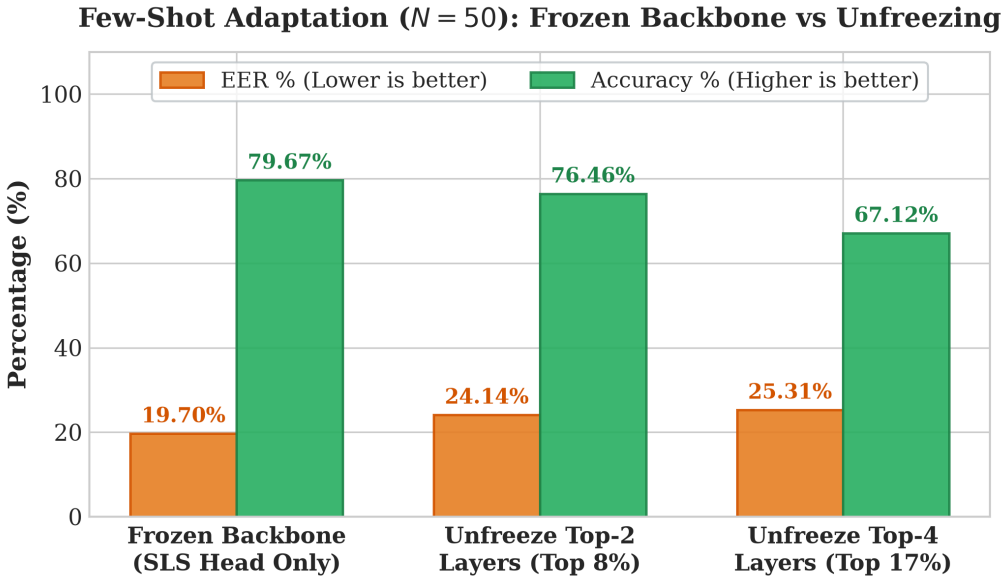


Figure 3: Ablation of Backbone Freezing vs. Partial Fine-Tuning. Unfreezing top transformer layers incurs substantial degradation in EER and generalization under low-data regimes due to representation collapse.

5.5 Generator Stratification & Detection Vulnerability

Figure 4 illustrates spoof detection accuracy across the five distinct text-to-speech architectures. While zero-shot transfer fails dramatically on neural diffusion models like `xtts-v2` and end-to-end models like `vits_mms`, few-shot adaptation at $N = 500$ achieves over 96% detection accuracy uniformly across all five generator paradigms.

6 Local Reproduction & Inference Protocols

6.1 Saved Model Weights Manifest

All target checkpoints have been downloaded from the GCP L4 instance into the local repository:

- `models/sls_head_n500_seed2_best.pth` (89 MB): Best performing few-shot model (98.00% Accuracy, 2.12% EER).
- `models/sls_head_n50_seed2.pth` (89 MB): Intermediate few-shot model ($N = 50$).
- `models/sls_head_english_pretrained.pth` (89 MB): Base zero-shot English pretrained checkpoint.

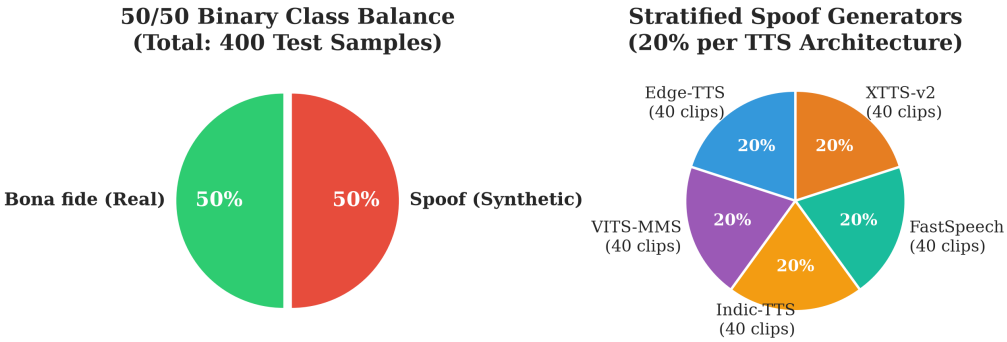


Figure 4: TTS Generator Stratification. Detection accuracy across Edge-TTS, FastSpeech, Indic-TTS, VITS-MMS, and XTTS-v2 across supervision levels.

6.2 Standalone Local Inference Execution

Inference can be executed locally on Apple Silicon (MPS) or CPU without cloud dependencies:

```
# Run inference on any arbitrary audio file using the N=500 best checkpoint:
python scripts/infer_local.py --audio sample.wav --device cpu

# Using Apple Silicon GPU acceleration:
python scripts/infer_local.py --audio sample.wav --device mps
```

References

[1] A. Babu, C. Wang, A. Tjandra, K. Lakhotia, Q. Xu, P. Goyal, K. Singh, P. von Platen, Y. Saraf, E. Pino, A. Baevski, A. Conneau, and M. Auli, "XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale," in *Proc. Interspeech*, 2022, pp. 2278–2282.

[2] Y. Zhang, S. Zheng, X. Mao, and L. Wang, "Sensitive Layer Summarization for Audio Deepfake Detection," in *Proc. ACM Multimedia*, 2024.